DEVELOPMENT OF A THERMOELECTRIC GENERATOR DEMONSTRATOR UNIT

“Advanced Radioisotope Mission Box”

TECHNICAL DISPLAY

Developed by:

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For:

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Introduction

The outlined project will detail the design, theoretical modeling, and hardware implementation of a Thermoelectric Generator (TEG) array designed to harvest thermal energy from a liquid medium. The primary objective of this development was to maximize energy conversion efficiency by optimizing the physical interface between the heat source and the Peltier modules.

This report documents the transition from an initial "coffee cup" conceptual design to a refined 4"x4"x4" 20-gauge steel cubic reservoir. This structural shift was necessitated by the need to eliminate the thermal gradients inherent in tangential contact with curved surfaces, thereby ensuring the bismuth telluride modules operate at their highest possible temperature differential.

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1. Basic Theory and Operation

1.1. The Seeback Effect

The fundamental principle utilized in this project is the Seebeck Effect, a phenomenon where a temperature difference between two dissimilar electrical conductors or semiconductors produces a voltage difference. In the ARMB, this is achieved using six TEC1-12706 bismuth telluride modules. When heat from the boiling water source is applied to one side of the module and the opposite side is exposed to a heat sink, a steady-state potential is generated. This process is the core mechanism enabling Radioisotope Thermoelectric Generators (RTGs) to power deep-space missions where solar energy is insufficient.

1.2. Thermoelectric Figure of Merit (Z)

The efficiency of the material is determined by the figure of merit, which combines the Seeback coefficient (α), electrical resistivity (ρ), and thermal conductivity (κ).

$$Z = \alpha^2/(\rho \cdot k) \tag{1}$$

Each module possesses an internal resistance of approximately 2.07 ohms [1]. For a series array (n=6) module, the total internal resistance is calculated by equations 2.

$$R_{total} = n \cdot R_{internal} \tag{2}$$

To achieve maximum power density, the load resistance should be matched to the internal resistance, whereas maximum efficiency requires an optimized ratio (*m'*). Equation 3 details this relationship.

$$R_L = m' \cdot R \tag{3}$$

The current produced by the temperature gradient across the array is calculated with Equation 4.

$$I = (\alpha \cdot \Delta T)/(R \cdot (m' + 1)) \tag{4}$$

While Z serves as the theoretical benchmark for the bismuth telluride modules, it is not plotted in the experimental results of this study. The current diagnostic suite is optimized for high-precision electrical output monitoring—specifically measuring voltage and current across the series array to determine power density.

Characterizing Z as a function of temperature would require synchronized, real-time temperature logging at both the hot-side steel interface and the cold-side heat sink for every data point. Because the current hardware configuration lacks an automated thermocouple array integrated into the microcontroller's data stream, the thermal variables required to solve for Z in real-time were not captured. Instead, this report focuses on the external performance metrics (V, I, and P) derived from the temperature gradient maintained by the 4"x4"x4" reservoir.

2. Plots

The presented plots are from the code running at a steady state input voltage, as such these plots are not physical under operational parameters. Presented plots are meant to show the code performing real time plotting of; the voltage, the current and the power coming from the temperature gradient.

Figure 1: Test run using power supply 5v input. Verified that code and circuit operating to design specifications.

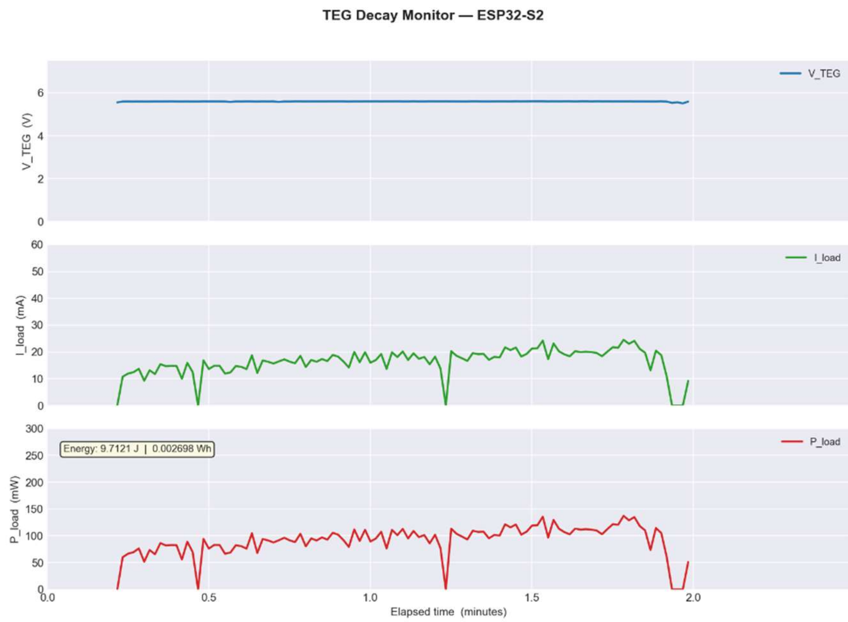
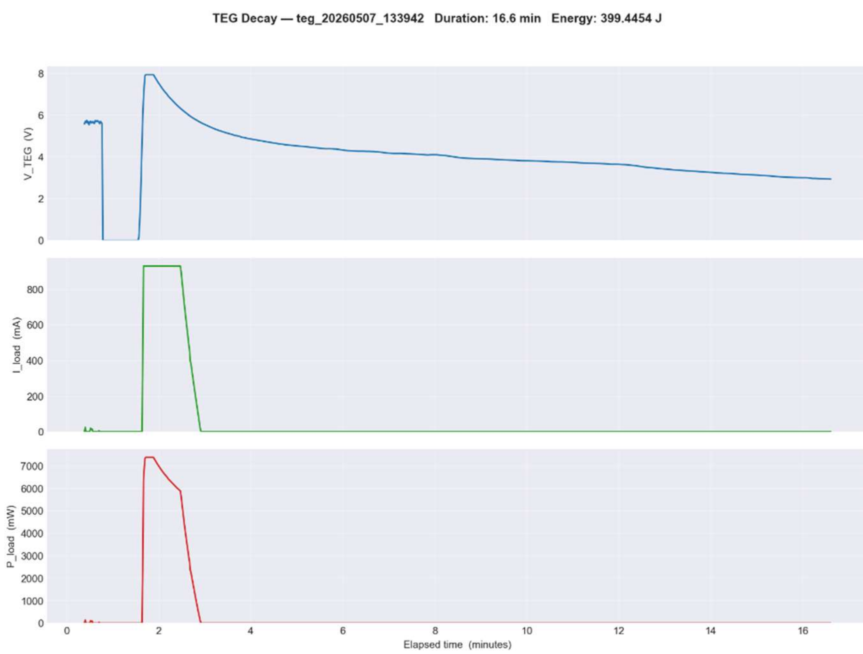


Figure 2: Real time plot of voltage during class presentation.



3. Brief Comments on Development

Initial ideas on what this project could look like assumed the form of a coffee cup surrounded by peltier modules. While practical and easily appealing to people, the practicality of attaching peltier modules that are flat to a curved surface proved to be a challenge. Initially an interface that would "bridge" this curve and the flat face of the modules was devised, however, this would necessitate the need for machining a metal "bridge". The idea being that metal has the highest thermal conductivity and would keep the temperature high. The second issue with this approach is the fact that the temperature experienced by the peltier module would be an average temperature. The center face of the module would experience the "truest" high temperature by the walls of the ceramic cup due to it being tangential. The surfaces right and left of the center would have thicker distances that heat would have to be transferred through. The idea of this can be seen in figure 3. As such, a redesign that had the peltier modules with as much contact with the medium as possible was preferred. A 4"x4"x4" 20 gauge steel cube was made and water sealed with commercial caulking. This yielded higher temperature gradients on the surface compared to the ceramic mug originally proposed when measured with an infrared thermometer.

Voltage measurements presented in the previous sections are those that are without uncertainties as the resistors used have a +/- 10% rating. Code that was made to display live voltage readings by way of a microcontroller. This presented challenges were to write the code and scale the TEG voltage to a usable voltage that the A/D on the micro controller could use. The input into the A/D was 0-3 volts.

Figure 3: X-Y cross section view of new design.

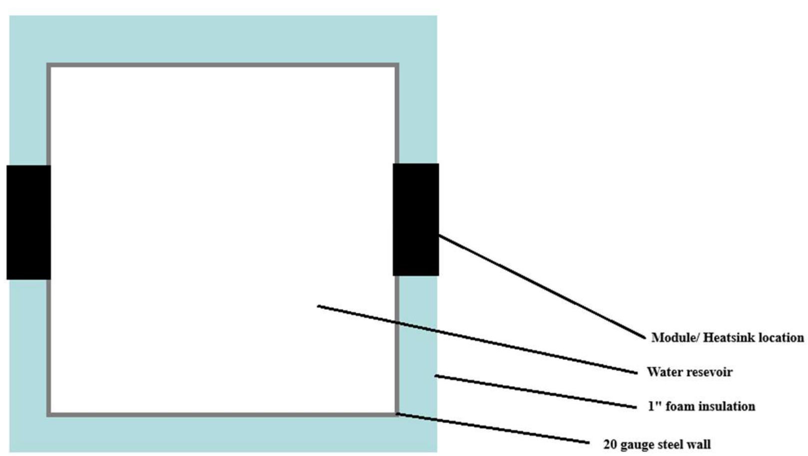
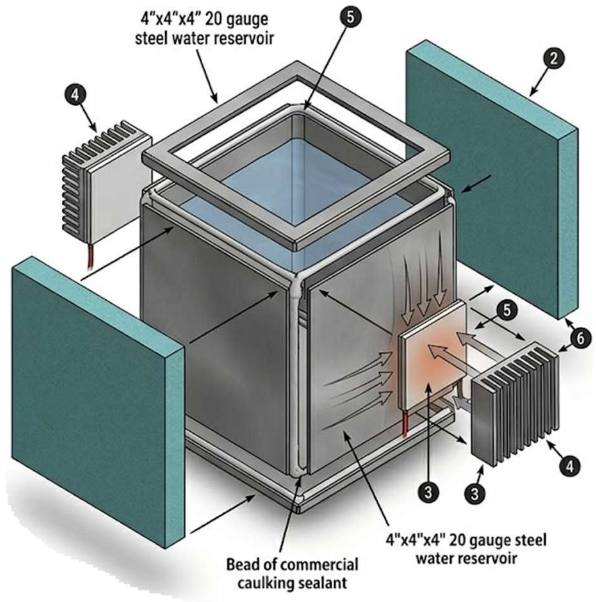


Figure 4: Exploded image of new design (<u>AI generated</u>; fed Figure 3 and detailed design characteristics. Missing heat sink fan details). Where 3,4 and 5 are related to the module and heatsink. 2 is the insulated foam and 5 is the caulking used to secure the edges.



4. Conclusion

The transition from a curved ceramic geometry to a cubic steel architecture successfully addressed the thermal transfer inefficiencies identified in the early stages of development. By utilizing flat 20-gauge steel walls, the Peltier modules achieved maximum surface contact, eliminating the need for complex "bridge" interfaces and ensuring the bismuth telluride modules were exposed to the truest possible temperature gradient. Infrared thermometry confirmed that this cubic design yielded significantly higher surface temperature differentials compared to the original coffee cup concept.

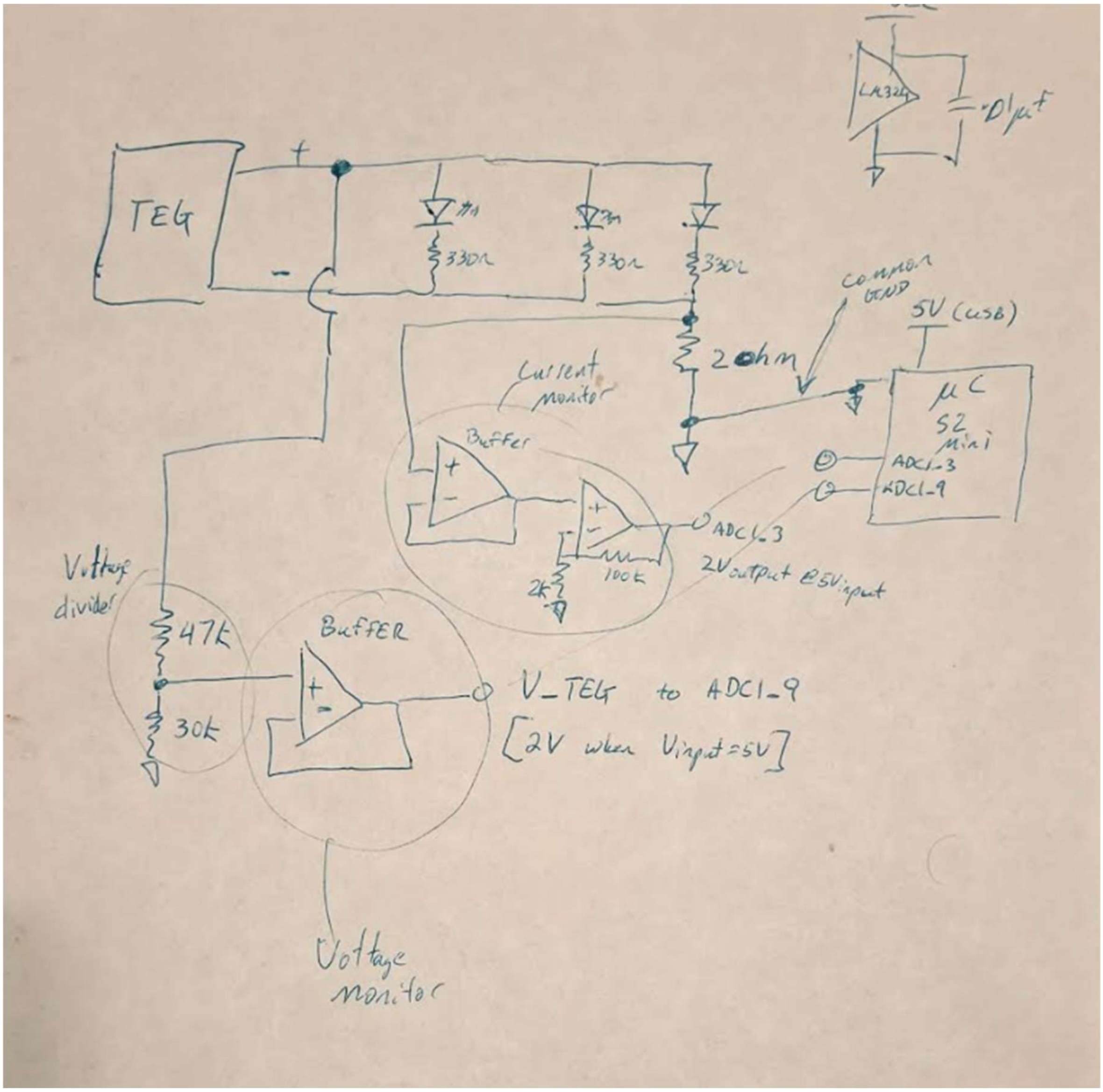
Furthermore, the integration of a microcontroller for real-time monitoring demonstrated the feasibility of tracking the Seebeck Effect in a controlled setting. While the +/- 10% tolerance of the resistors introduced minor uncertainties in the voltage scaling for the 0-3V A/D input, the system effectively displayed the core relationships between temperature gradients and power output. This project successfully validates the fundamental mechanics of solid-state thermal energy harvesting, providing a scalable model for the same thermoelectric principles that power critical deep-space instrumentation.

Authors Note:

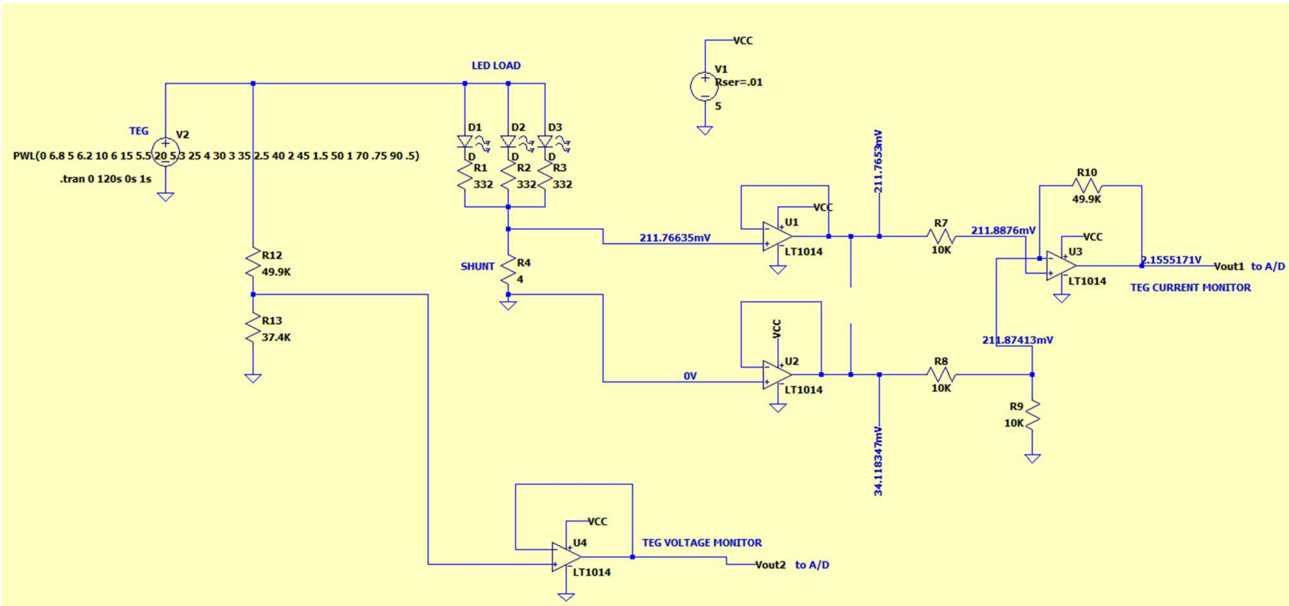
Presented circuit for LTSpice had shunt placed where the high voltage was, this was moved to the low side as seen in the physical drawn circuit. It was decided to include the simulated circuit for completeness of data and to show the decline of voltage over time. Inputs for voltage as a function of time was input into LTSpice to give physical voltage readings taken from a Fluke 189 Multimeter.

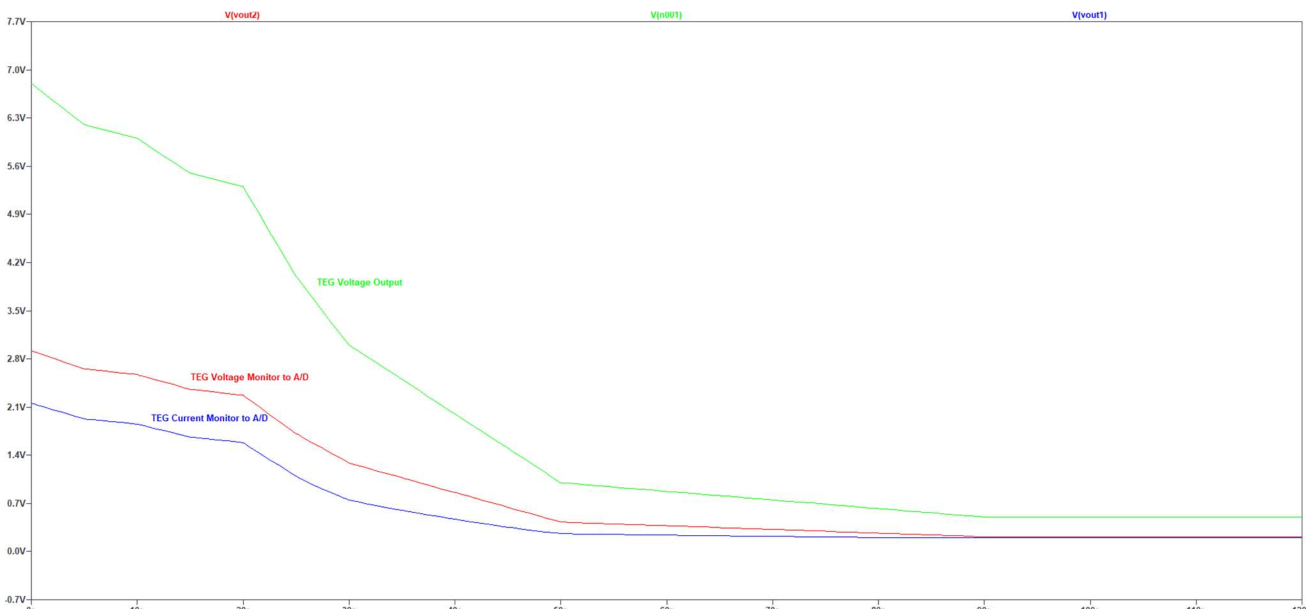
5. Appendices

a. Current Circuit Used



b. Simulated Circuit (LTSpice – Iteration 1)





c. Physical Images

d. Voltage Slopes

Time Interval (min)	Voltage Slope (V/min)	Intercept (V)	R^2 Value
8 – 12 min	-0.10096	4.8353	0.9630
12 – 16 min	-0.15887	5.5004	0.9842

The presented slopes in this section are from the in-class demonstration. The slopes presented differ in the fact that the 12–16-minute time period has the lip opened on the module to allow for convective cooling. The 8-to-12-minute time period was the stable slope period on the decline.

6. References

[1] – NE568 Introduction to Space Nuclear Power, Muhamed El Genk. 2026